

Joshua Li

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EDUCATION

Arizona State University | Tempe, AZ

Masters of Science, Computer Science

Graduating: Dec 2026, GPA: 4.0

University of California, San Diego | San Diego, CA

Bachelor of Science, Data Science

Graduated: Mar 2025, GPA: 3.9

WORK EXPERIENCE

BILL | San Jose, CA

Associate Fraud Risk Strategy Data Scientist

Dec 2025 - Present

- Engineered a sequential adaptive-floor beam search to automate fraud rule discovery across three product lines, reducing manual referrals by 21% while increasing fraud detection recall to 99.3% and capturing \$83K in additional fraudulent transactions.
- Developed a NetworkX identity graph on AWS SageMaker processing 90K organizations and 91K edges across 21 link vectors, achieving 93.5% precision and 99.4% recall via weekly automated retraining, which linked 231 of 1,645 fraud organizations to prohibited entities, identifying \$213K of \$660K of fraud in FY27 as preventable.
- Deployed daily production monitoring agents as scheduled SageMaker jobs over Starburst with Slack integration, leveraging Z-score anomaly detection with seasonal controls for dispute tracking and organization performance visibility.

Single Particle LLC | San Diego, CA

Software Programming Intern

Jul 2023 - Oct 2023

- Fine-tuned ChatGPT models for cryo-electron microscopy (cryoEM) workstations and led the integration of AI into the company website.

Intel Corporation | Folsom, CA

Software Programming Intern

Jun 2022 - Sep 2022

- Engineered and rigorously tested data science workflows tailored to various industry sectors, enhancing team insights into workstation performance.
- Conducted extensive benchmarking of workflows by simulating real-world industry use cases using Python and Scikit-Learn, improving efficiency by 20%.

PROJECTS

Optimizing Training Cost for Scalable Graph Processing | UCSD, Qualcomm

Python, PyTorch, NetworkX, CUDA

- Worked closely with Qualcomm to develop a scalable graph-based learning pipeline to predict congestion in chip designs using the DE-HNN model.
- Processed 6 large-scale netlists (460k to 920k nodes) into bipartite graph structures, integrating spectral and structural features for demand prediction.
- Achieved 89.3% runtime reduction and 38.8% peak GPU memory savings.
- Delivered a final model with only a 6% average performance drop while dramatically lowering compute costs.

SKILLS

- Languages:** Python, SQL, R, C, C++, Java, JavaScript, TypeScript, HTML/CSS, MATLAB
- Frameworks & Libraries:** Pandas, PyTorch, TensorFlow, Keras, Dask, Scikit-learn, NetworkX, Plotly, Seaborn, React, D3, Flask, Django, Spark, Hadoop, Optuna
- Databases & Big Data:** MySQL, Postgres, NoSQL, MongoDB, Starburst (Trino), Databricks
- Tools & Platforms:** AWS (Sagemaker, S3), Google Cloud, Docker, Tableau, Git, GitHub, JSON, Microsoft Excel, Node.js, Data Visor, LiteLLM, Ollama